WMS-AI

Project Report

Warehouse Management System with Intelligent Recommendations

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1. Topic

WMS-AI is a web-based Warehouse Management System created with Java Spring Boot and React.js. It manages inventory operations and provides smart recommendations using a blend of rule-based logic and AI-based suggestions.

2. Abstract

Traditional warehouse management systems mainly focus on storing and retrieving inventory data. These systems do not support proactive decision-making or offer intelligent insights.

WMS-AI improves on this by adding automated alerts and an intelligent recommendation engine. It continuously monitors stock levels and generates alerts when predefined thresholds are exceeded. It also suggests product bundling and strategies to handle overstock situations.

This application uses Spring Boot for the backend, React.js for the frontend, and PostgreSQL for data storage. It incorporates JWT-based authentication and role-based access control. Additional features include analytics dashboards, email notifications, and functionality for exporting and importing files.

3. Problem Statement

Warehouse management systems typically face the following problems:

- No real-time monitoring of stock levels

- Inability to effectively detect overstock situations

- Lack of smart recommendations for increasing sales

WMS-AI aims to solve these problems by introducing automation, alert systems, and suggestion features.

4. Objectives

- To create a complete warehouse management system
- To automate inventory monitoring and alert generation
- To offer smart recommendations through a hybrid approach
- To support data-driven decision-making with analytics
- To use object-oriented programming concepts in a practical system

5. Functional Requirements

5.1 Authentication Module

- Users must log in with their email and password
- Passwords must be stored in a hashed format
- JWT-based authentication must be active
- Role-based access control must be in place

5.2 Product and Inventory Management

- The system must allow users to create, update, delete, and search for products
- Each product must have attributes like SKU, category, supplier, price, and quantity
- The system must support stock-in and stock-out operations
- Stock-out must check available quantity before processing
- Alerts must trigger when thresholds are exceeded

5.3 Supplier Management

- The system must let users create and manage suppliers
- Suppliers must be searchable by name and ID
- The system must display products linked to each supplier

5.4 Order Management

- The system must allow the creation of purchase orders
- Order status must be trackable (Pending, Sent, Received, Canceled)

- Stock must automatically update when orders are received

5.5 Sales Management

- The system must record sales transactions
- Inventory must adjust after each sale
- Transaction logs must be maintained

5.6 Alert and Notification System

- The system must generate alerts for low stock and overstock conditions
- Email notifications must be sent to the administrator
- Alerts must appear on the dashboard
- Duplicate alerts must be blocked within a set time period

5.7 Recommendation Engine

- The system must suggest product bundles for new products
- The system must offer strategies for managing overstock
- Recommendations must come from a hybrid approach:
  - Rule-based logic
  - AI-based suggestions

5.8 Analytics Module

- The system must display sales and inventory statistics
- The system must support comparisons based on time periods (MoM, QoQ, YoY)
- The system must identify top-selling and slow-moving products

5.9 File Handling

- The system must export reports in CSV and TXT formats
- The system must support product imports using CSV files
- The system must maintain logs and backups

6. Non-Functional Requirements

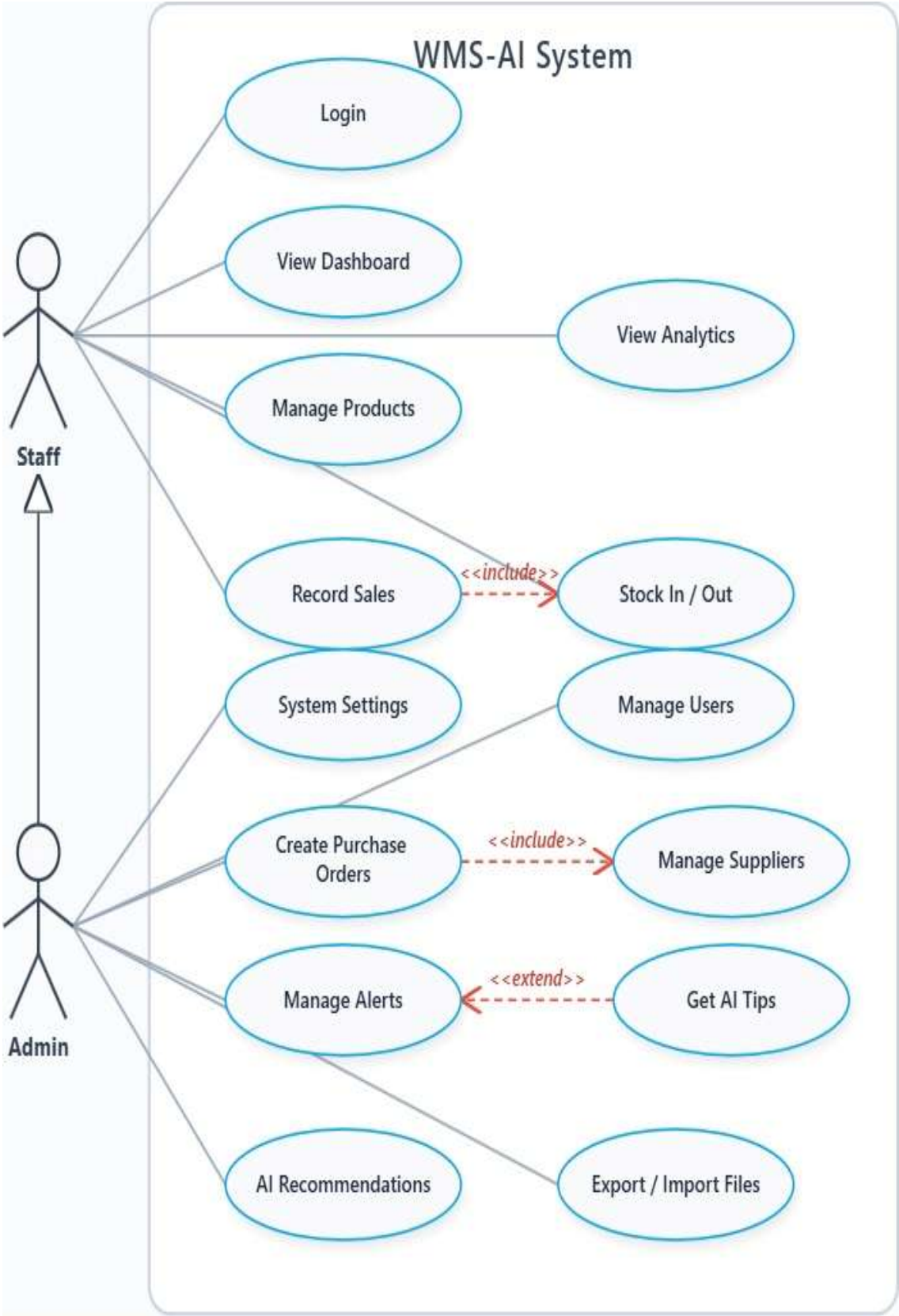
- The system must provide response times under 500 milliseconds for standard operations
- Data must be stored securely
- The system must be maintainable and scalable
- The system must work across different environments
- Proper error handling and logging must be in place

7. System Architecture

The system follows a layered architecture consisting of:

- Presentation Layer (React.js)
- Application Layer (Spring Boot REST API)
- Service Layer (Business logic)
- Data Layer (PostgreSQL database)
- External AI Layer (Recommendation API)

8. Use Case Diagram



9. Key Features

- Inventory management
- Automated alerts
- AI-based recommendations
- Dashboard analytics
- Role-based authentication
- File export and import

10. Technologies Used

Backend

- Java
- Spring Boot
- Spring Security
- Hibernate (JPA)

Frontend

- React.js
- Tailwind CSS

Database

- PostgreSQL

Additional Tools

- Gmail SMTP
- AI API

11. Object-Oriented Concepts

- Encapsulation through private fields and access methods
- Inheritance through class hierarchy (Person → Employee)
- Polymorphism through method overloading
- Abstraction through interfaces
- Exception handling using custom exceptions
- Collections such as ArrayList and PriorityQueue

12. Conclusion

WMS-AI is a structured warehouse management system that combines standard inventory operations with smart features. The system boosts efficiency by automating monitoring and offering actionable recommendations. It also showcases the use of object-oriented programming concepts in a practical application.